

Department of Computer Science
CS403NOD – Node.js

Lab Assignment – 02
Building Your First Node.js Server
Semester: BCA VII

Objective

After completing this lab, you should be able to:

- Understand how Node.js handles requests using the core http module.
- Create and run a basic HTTP server.
- Handle multiple routes within a single server.
- Send different types of responses (plain text and JSON).
- Understand the request and response objects.
- Continue pushing project updates to GitHub.

Tasks

Task 1 – Create Project Folder

Create the following folder structure.

```
NodeJS-Lab/  
└── Lab-02/  
    ├── server.js  
    ├── package.json  
    ├── README.md  
    ├── server-running.png  
    ├── routes-output.png  
    └── architecture-notes.txt
```

Task 2 – Initialize Node Project

Open Terminal inside Lab-02.

Run

```
npm init -y
```

This should create a package.json file.

Task 3 – Create Your First Server

Create

```
server.js
```

Write a server using the http module that listens on port 3000 and responds on the home route (/) with a welcome message containing:

- Your Name
- Scholar Number
- Course

Run the server.

```
node server.js
```

Open the following address in your browser.

```
http://localhost:3000
```

Take a screenshot.

Save it as

```
server-running.png
```

Task 4 – Add Multiple Routes

Add two more routes to the same server.

- /about – returns a short message about yourself
- /college – returns your college name and semester

Test all three routes (/ , /about , /college) in the browser.

Take a screenshot showing the output of all three.

Save it as

routes-output.png

Task 5 – Send a JSON Response

Add a route

/profile

that returns a JSON object containing name, scholarNumber, course, semester, and college.

Set the response header Content-Type to application/json before sending it.

Task 6 – Handle Unknown Routes (404)

Add a final else block to your server that returns status code 404 with the message “Page Not Found” for any route that is not handled above.

Task 7 – Use an Environment Variable for the Port

Modify server.js to read the port from an environment variable instead of a fixed number.

const PORT = process.env.PORT || 3000;

Use PORT inside server.listen() and confirm the server still runs correctly.

Task 8 – Understanding Node.js Architecture

Create a text file named

architecture-notes.txt

Answer the following in your own words (4–5 sentences each):

- What is the Event Loop?
- Is Node.js single-threaded or multi-threaded? Explain.
- What is the role of libuv?

README.md

Update the README.md you created in Lab 01 to also include:

- A list of every route in your server and what each one returns
- Lab Number: 02
- Date

GitHub Submission

Your repository already exists from Lab 01, so you do not need to run git init again.

Step 1

Open Terminal inside NodeJS-Lab (the main project folder, not just Lab-02).

Step 2

Run the following commands.

git add .

git commit -m "Lab 02 Submission"

git push

If your repository is not yet connected for some reason, run these first.

git remote add origin <repository-url>

git branch -M main

git push -u origin main

Google Classroom Submission

Submit

- GitHub Repository Link
- (Optional) ZIP File

If You Face Any Problem

If you are unable to complete any task, **do not leave it incomplete without explanation.**

Create a section in your **README.md** named:

Problems Faced

Mention:

- Task Number
- Error / Issue
- What you tried to solve it
- Screenshot (if applicable)

Example

Problems Faced

Task No.: Task 5

Issue:

Browser shows [object Object] instead of the actual JSON data.

Attempted Solution:

Used `JSON.stringify()` on the object before passing it to `res.end()`, which fixed it.

Students who clearly document genuine problems will receive guidance during the next lab session.